IRBESARTAN (AVAPRO)

Pharma Talk



Antihypertensive | Angiotensin II Receptor Blocker (ARB)

Blocks AT1 receptor to lower blood pressure and protect kidneys.

1 MECHANISM OF ACTION (MOA) & TARGET



- TARGET: Angiotensin II type 1 (AT1) receptor
- MOA: Blocks AT1 binding → ↓ aldosterone secretion & vasodilation → ↓ BP
- Analogy: A shield blocking the vasoconstricting hormone from gripping blood vessels.

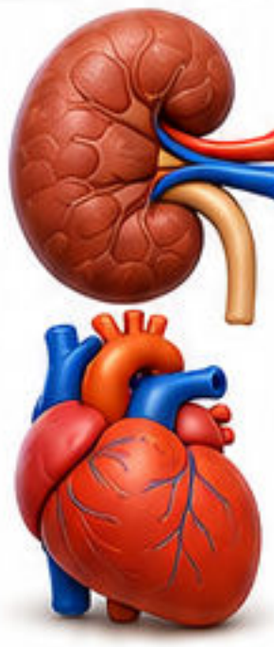
2 FORMS & STRENGTHS

- Tablet: 75 mg, 150 mg, 300 mg



75 mg | 150 mg | 300 mg

3 INDICATIONS (FDA-APPROVED & MAJOR OFF-LABEL)



- **FDA-APPROVED:**
  - Hypertension (HTN)
  - Diabetic nephropathy in T2DM with elevated serum creatinine & proteinuria.
- **MAJOR OFF-LABEL:**
  - Heart failure with reduced ejection fraction (HFrEF) in ACEi-intolerant patients.

4 DOSAGE & ADMINISTRATION

A) ADULT DOSE

- HTN: 150 mg PO daily
- Max: 300 mg/day
- Nephropathy: Target 300 mg PO daily

B) PEDIATRIC DOSE

- PEDIATRIC USE — Safety/efficacy not established.

C) ADMINISTRATION



- Route: Orally
- Take with or without food

D) ORGAN ADJUSTMENT



- Renal: No adjustment
- Hepatic: No adjustment for mild/moderate

5 PHARMACOKINETICS & DYNAMICS (PK/PD)



- Onset of action: 1–2 hours
- Half-life ($T_{1/2}$): 11–15 hours
- Metabolism: Hepatic (CYP2C9)

6 ADVERSE EFFECTS & TOXICITY

COMMON AEs

- Dizziness
- Fatigue
- Orthostatic hypotension



SEVERE / SERIOUS AEs

- Hyperkalemia
- Angioedema
- Acute kidney injury (AKI)



7 CONTRAINDICATIONS & INTERACTIONS

ABSOLUTE CONTRAINDICATION

- Do NOT use with aliskiren in patients with diabetes.

BLACK BOX WARNING
FETAL TOXICITY
Can cause fetal injury and death when used in pregnancy.

MAJOR DRUG INTERACTIONS

- Lithium → ↓ excretion → ↑ lithium levels → toxicity
- NSAIDs (e.g., ibuprofen) → ↓ renal perfusion → ↑ risk of AKI

8 SPECIAL POPULATIONS (PREGNANCY & LACTATION)



Pregnancy: Contraindicated



Lactation: Avoid use



Geriatric: Start at 75 mg if volume-depleted.

9 CLINICAL PEARLS



1. Superior to amlodipine for slowing diabetic nephropathy progression.
2. Monitor K^+ and SCr 1–2 weeks post-initiation.
3. Do not combine with ACEi due to ↑ hyperkalemia & AKI risk.
4. Counsel women of childbearing age on strict contraception.
5. Avoid in bilateral renal artery stenosis.

10 GLOSSARY OF ABBREVIATIONS

ARB	= Angiotensin II Receptor Blocker	CYP	= Cytochrome P450
AT1	= Angiotensin II type 1	AKI	= Acute Kidney Injury
T2DM	= Type 2 Diabetes Mellitus	DM	= Diabetes Mellitus
HFrEF	= Heart Failure with Reduced Ejection Fraction	SCr	= Serum Creatinine
ACEi	= Angiotensin-Converting Enzyme Inhibitor		